Grover Aruquipa

Ph.D. Candidate in Mechanical Engineering • Université Laval, Québec, Canada

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Research Profile

I design, build and control robots whose capabilities come from the mechanism itself: parallel robots with kinematic redundancy, unlimited rotations and grasping produced by the structure rather than by an added end-effector. My work runs from type synthesis, kinematics, statics and singularity analysis, through simulation and mechanical design, to fabricated prototypes driven by real-time multi-motor control.

Output at a glance. 3 journal articles and 3 refereed conference papers published, first author on all; 4 papers under review (3 journals, 1 ICRA); 3 prototypes designed, built and experimentally validated, a fourth under construction; 2 graduate students supervised day to day. First place, ASME Student Mechanisms and Robotics Design Competition (2025); finalist, MassRobotics Form & Function Challenge (2026).

Education

Ph.D., Mechanical Engineering 2023 – Jan. 2027 (expected)

Université Laval, Québec, Canada. Supervisor: Prof. Clément Gosselin; co-director: Prof. Patrice Lambert.

Dissertation: *Kinematically redundant parallel robots with articulated platforms for unlimited rotation and grasping.*

M.Sc., Ingénierie des systèmes complexes, Smart Integrated Systems 2021 – 2023

Université Bourgogne–Franche-Comté (UBFC), Besançon, France. Mention Bien; 120 ECTS. Supervisor: Prof. Redwan Dahmouche. Research thesis with FEMTO-ST (CNRS UMR 6174).

B.Eng., Mechatronics Engineering 2014 – 2019

Bolivian Catholic University, La Paz, Bolivia. Thesis on a reconfigurable Delta-type parallel robot for product picking.

Awards, Distinctions and Scholarships

1st Place, Graduate Division, ASME Student Mechanisms and Robotics Design Competition (SMRDC) 2025

ASME IDETC/CIE, Anaheim, 17–20 Aug. 2025. Team led by me; entered as *The First Parallel Robot with Two Unlimited Rotations*.

Finalist, MassRobotics Form & Function Challenge (Boston; three-rotation prototype) 2026

Mitacs Globalink Research Award (GRA) 2026

Mitacs, Canada; competitive award funding a research stay at JKU Linz with Prof. A. Müller.

1st Place, Student Paper Competition, X Jornadas Argentinas de Robótica (JAR) 2019

Universidad Nacional del Comahue, Neuquén, Argentina.

Doctoral and master's funding 2021–2026

Université Laval Ph.D. Scholarship and Citoyens du Monde Grant (2023–2026) • Screw Theory Summer School, TU Delft (2023) • UBFC Master's Scholarship (2021–2023).

Publications

Author order reflects contribution; my name is **G. Aruquipa**. Q1 denotes journal quartile (Scimago).

Peer-reviewed journal articles (published)

[J1] **G. Aruquipa**, P. Lambert, R. Dahmouche, C. Gosselin, "Parallel robots for grasping tasks: a review," *ASME Journal of Mechanisms and Robotics* Q1, vol. 18, no. 12, art. 120801, pp. 1–26, 2026. doi:10.1115/1.4072612

[J2] **G. Aruquipa**, V. Muralidharan, P. Lambert, C. Gosselin, "Analysis and design of a novel compact four-degree-of-freedom parallel robot with unlimited rotation," *ASME Journal of Mechanisms and Robotics* Q1, vol. 18, no. 6, art. 061001, 2026.

[J3] **G. Aruquipa**, P. Lambert, C. Gosselin, "Analysis and design of a novel (6+1)-DoF parallel robot with unlimited rotation and grasping capabilities," *Mechanism and Machine Theory* Q1, vol. 224, art. 106438, 2026.

Refereed conference proceedings

[C1] **G. Aruquipa**, P. Lambert, C. Gosselin, "Kinematic analysis and design of a novel 9-DoF parallel robot with grasping capabilities," *CCToMM Symposium on Mechanisms, Machines and Mechatronics*, 2025, pp. 117–127. doi:10.1007/978-3-031-95489-4_10

[C2] **G. Aruquipa**, F. Diaz, "An IoT architecture based on the control of a bio-inspired manufacturing system for the detection of anomalies with vibration sensors," *Procedia Computer Science*, vol. 200, pp. 438–450, 2022. doi:10.1016/j.procs.2022.01.242

[C3] **G. Aruquipa**, G. A. Rojas Silva, "Analysis of algorithmic trading with Q-learning in the Forex market," *IEEE Int. Conf. on Emerging Smart Computing and Informatics (ESCI)*, 2021, pp. 73–77. Indexed as A. A. Grover, R. S. Gabriel. doi:10.1109/ESCI50559.2021.9396948

Under review

[S1] **G. Aruquipa**, P. Lambert, C. Gosselin, "An analytical framework for collision-free and singularity-free unlimited rotation in spatial parallel mechanisms," *Mechanism and Machine Theory* Q1.

[S2] **G. Aruquipa**, P. Lambert, C. Gosselin, "A compact coaxial parallel wrist-gripper with unlimited tool-axis rotation," *Robotics and Autonomous Systems* (Elsevier) Q1.

[S3] **G. Aruquipa**, A. Ayoub, R. Dahmouche, "Architecture, kinematics, and workspace analysis of a redundant 8RSS parallel robot with two unlimited rotations," *Mechanism and Machine Theory* Q1.

[S4] **G. Aruquipa**, P. Lambert, C. Gosselin, "A kinematically redundant (6+2)-DoF parallel robot with three orthogonal unlimited rotations and coaxial actuation," *IEEE International Conference on Robotics and Automation (ICRA 2027)*.

Manuscripts in preparation

G. Aruquipa, A. Müller, P. Lambert, C. Gosselin, reduced graspability Jacobian for parallel robots with rotationally articulated platforms (in co-author review); a four-leg Delta robot with an integrated Sarrus gripper and grasp-force estimation from motor currents (prototype under construction).

Prototypes and Experimental Systems

Coaxial parallel wrist-gripper, 4 DoF 2025–2026

Four actuation channels through concentric hollow shafts, 180 mm footprint, 0.25 kg moving mass; unlimited tool-axis rotation with a non-backdrivable gripper. Real-time control at 500 Hz, validated over ten continuous revolutions with sub-degree tracking.

(6+1)-DoF redundant parallel robot with integrated grasping 2024–2025

Seven base-mounted actuators; gripper driven from the base through a concentric tube.

(6+2)-DoF parallel robot with three orthogonal unlimited rotations 2024–2026

Open-loop reconfigurable platform for grasping; built with the two graduate students I supervise.

Four-leg Delta robot with integrated Sarrus gripper 2026–present

SLS-printed links, carbon-tube forearms, belt or direct drive; dynamic model validated in Simscape before hardware tests; grasp force estimated from motor currents to handle fragile objects at speed.

Research and Professional Experience

Visiting Researcher, Institute of Robotics, Johannes Kepler University Linz Apr.–May 2026

Linz, Austria. Host: Univ.-Prof. Andreas Müller. Singularity analysis and graspability performance indices for parallel robots with integrated grasping; funded by the Mitacs Globalink Research Award.

Doctoral Researcher, Robotics Laboratory, Université Laval 2023–2027

Type synthesis, kinematic and singularity analysis, mechanical design, fabrication and real-time control of redundant parallel robots with unlimited rotation and integrated grasping.

Research Intern, FEMTO-ST (CNRS UMR 6174), Besançon, France 2022–2023

Master's research on a redundant dexterous parallel robot for micro-manipulation; real-time C++ control software.

Industry and data science: Hitachi ABB Power Grids R&D, Zurich; Sencon Srl; KTH (Sweden) 2020–2021

Supervision, Service and Outreach

Co-supervision of graduate research 2025–2026

Day-to-day supervision of two graduate students, Brandon Valencia and Alireza Hobubaty, on the (6+2)-DoF redundant architecture, leading to the manuscript submitted to ICRA 2027.

Reviewer, *ASME Journal of Mechanisms and Robotics* 2026–present

Outreach and mentoring, Bolivia 2016–present

Lectures on parallel robots at Bolivian universities; scholarship mentor, TuBeca Bolivia; volunteer, HOPE Worldwide.

Technical Skills

Modelling and analysis: screw theory, Lie group methods, type synthesis, statics, singularity and workspace analysis; MATLAB. **Simulation:** MuJoCo, Simscape (dynamic models validated against hardware). **Motor control and real time:** multi-motor pipelines at 500 Hz with incremental encoders, OPAL-RT RT-LAB, Simulink Real-Time, real-time C++. **Robotics software:** ROS and ROS 2, MoveIt, Pinocchio, OpenCV, Python. **Design and fabrication:** SolidWorks, Altium; machining, SLS and FDM printing, wiring and assembly of prototypes. **Machine learning:** PyTorch. **Languages:** Spanish (native), English (full professional), French (intermediate).

References

Prof. Clément Gosselin, Full Professor, Université Laval (Ph.D. supervisor), clement.gosselin@gmc.ulaval.ca

Univ.-Prof. Dr.-Ing. Andreas Müller, Head, Institute of Robotics, JKU Linz (host of my 2026 research stay), a.mueller@jku.at

Prof. Patrice Lambert, Assistant Professor, Université Laval (Ph.D. co-supervisor), patrice.lambert@gmc.ulaval.ca